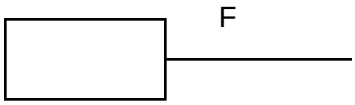
GMF authored QA fixture - CC0 - NOT an official AP paper

Section I: Multiple Choice

Questions 1-2 refer to the following block diagram.



1. A block has mass 2 kg. Use net F = ma. Find acceleration for 6 N.

(A) 1 m/s^2

(B) 3 m/s^2

(C) 6 m/s^2

2. The same block receives 8 N. Use net F = ma.

(A) 2 m/s^2

(B) 4 m/s^2

(C) 8 m/s^2

Section II: Free Response

1. Derive an expression for the acceleration.

(A) Explain the force model.

(B) Explain the dimensions.

Answer Key

1 B 2 B